

99.500 Applied Mathematics for Engineering Spring Semester AY2026

Separation of Variables

Week 9 Outline

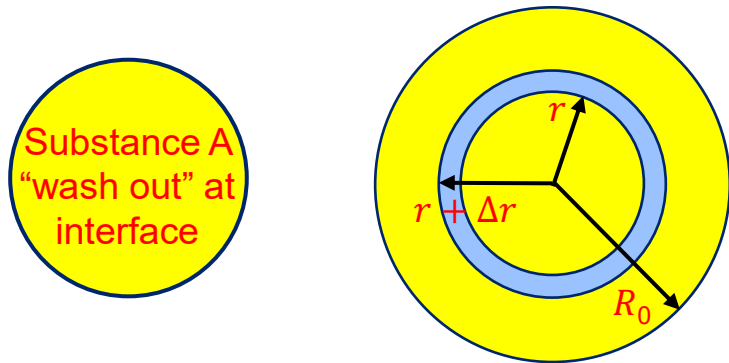
- Non-steady-state “spherical” diffusion problem
- Sturm-Louiville System
- Example of diffusion-only transport
- Example of heat conduction problem

Non-steady-state diffusion in a spherical capsule

Bessel ODEs naturally arise in problem with radial symmetry

Example: Diffusive reaction in a spherical shaped catalyst

Concentration of A: $C = C(r, t)$



$$\begin{aligned} \text{In:} & \left(-D \frac{\partial C}{\partial r} 4\pi r^2 \right) \Big|_r \\ \text{Out:} & \left(-D \frac{\partial C}{\partial r} 4\pi r^2 \right) \Big|_{r+\Delta r} \end{aligned}$$

Generation: 0

$$\text{Accumulation: } \frac{\partial(C\Delta V)}{\partial t} = 4\pi r^2 \Delta r \frac{\partial C}{\partial t}$$

Mass balance for substance A:

$$4\pi r^2 \Delta r \frac{\partial C}{\partial t} = \left(-D \frac{\partial C}{\partial r} 4\pi r^2 \right) \Big|_r - \left(-D \frac{\partial C}{\partial r} 4\pi r^2 \right) \Big|_{r+\Delta r} \implies \frac{\partial C}{\partial t} = \frac{D}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C}{\partial r} \right)$$

Initial condition: $C(r, t = 0) = C_0$

Initial condition: $C(r, t = 0) = C_0$

$$\text{Boundary conditions: } \begin{cases} \frac{\partial C}{\partial r} \Big|_{r=0} = 0 \\ C(R_0, t) = 0 \end{cases}$$

Separation of Variables

$$\frac{\partial C}{\partial t} = \frac{D}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C}{\partial r} \right)$$

$$IC : C(r, t = 0) = C_0$$

$$BC1 : \left. \frac{\partial C}{\partial r} \right|_{r=0} = 0$$

$$BC2 : C(R_0, t) = 0$$

Assume: $C(r, t) = R(r)T(t)$ (Separable assumption)

Substitute into the PDE and boundary conditions:

$$\frac{\partial(RT)}{\partial t} = \frac{D}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial(RT)}{\partial r} \right) \implies R \frac{dT}{dt} = T \frac{D}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right)$$

$$\implies \frac{1}{T} \frac{dT}{dt} = \frac{1}{R} \frac{D}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right)$$

$$\frac{1}{DT} \frac{dT}{dt} = \frac{1}{R} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\lambda^2$$

Time-Dependent vs Radial Part

Solve for the time-dependent part first:

$$\frac{1}{DT} \frac{dT}{dt} = -\lambda^2 \implies \frac{dT}{T} = -D\lambda^2 dt \implies T = Ae^{-\lambda^2 Dt}$$

Solve for the radial part:

$$\begin{aligned} \frac{1}{R} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) &= -\lambda^2 \implies \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\lambda^2 R r^2 \\ &\implies r^2 R'' + 2r R' + \lambda^2 r^2 R = 0 \\ \text{Generalized Bessel ODE} &\implies R = \frac{1}{\sqrt{r}} \left[b_1 J_{\frac{1}{2}}(\lambda r) + b_2 J_{-\frac{1}{2}}(\lambda r) \right] \\ &= \frac{1}{r} [c_1 \cos(\lambda r) + c_2 \sin(\lambda r)] \end{aligned}$$

Generalized Bessel ODE

$$x^2 y'' + x(a + 2bx^r)y' + [c + dx^{2s} - b(1 - a - r)x^r + b^2 x^{2r}] y = 0; \quad \text{for } r, s \neq 0$$

with general solution in the form

$$y = x^{\frac{1-a}{2}} e^{-\frac{bx^r}{r}} \left[AZ_p \left(\frac{\sqrt{|d|}}{s} x^s \right) + BZ_{-p} \left(\frac{\sqrt{|d|}}{s} x^s \right) \right] \quad \text{where } p = \frac{1}{s} \sqrt{\left(\frac{1-a}{2} \right)^2 - c}$$

1. If $\frac{\sqrt{d}}{s}$ is *real* and p is *not* an integer (or zero), then Z_p denotes J_p and Z_{-p} denotes J_{-p}
2. If $\frac{\sqrt{d}}{s}$ is *real* and p is zero or an integer k , then Z_p denotes J_k and Z_{-p} denotes Y_k
3. If $\frac{\sqrt{d}}{s}$ is *imaginary* and p is *not* zero or an integer, then Z_p denotes I_p and Z_{-p} denotes I_{-p}
4. If $\frac{\sqrt{d}}{s}$ is *imaginary* and p is zero or an integer k , then Z_p denotes I_k and Z_{-p} denotes K_k

Apply Boundary Conditions

$$BC1: \left. \frac{\partial C}{\partial r} \right|_{r=0} = 0 \implies T(t) \left. \frac{dR}{dr} \right|_{r=0} = 0 \implies R'(0) = 0$$

$$\text{finite solution at } R(0) \implies c_1 = 0 \implies R = c_2 \frac{\sin(\lambda r)}{r}$$

$$BC2: C(R_0, t) = 0 \implies R(R_0)T(t) = 0 \implies R(R_0) = 0$$

$$\implies R(R_0) = c_2 \frac{\sin(\lambda R_0)}{R_0} = 0$$

$$\implies c_2 = 0 \text{ or } \sin(\lambda R_0) = 0$$

$$\sin(\lambda R_0) = 0 \implies \lambda R_0 = n\pi, n = 1, 2, \dots \implies \lambda_n = \frac{n\pi}{R_0}$$

$$\begin{cases} T_n = A_n e^{-\lambda_n^2 D t} \\ R_n = c_n \frac{\sin(\lambda_n r)}{r} \end{cases} \quad \lambda_n = \frac{n\pi}{R_0}, n = 1, 2, \dots$$

$$C(r, t) = \sum_{n=1}^{\infty} R_n(r) T_n(t) = \sum_{n=1}^{\infty} B_n \frac{\sin(\lambda_n r)}{r} e^{-\lambda_n^2 D t}$$

How to
determine B_n ?

$$C(r, t=0) = C_0 \implies \sum_{n=1}^{\infty} B_n \frac{\sin(\lambda_n r)}{r} e^{-\lambda_n^2 D \cdot 0} = \sum_{n=1}^{\infty} B_n \frac{\sin(\lambda_n r)}{r} = C_0$$

Sturm-Liouville (S-L) System

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x)y = -f(\lambda)w(x)y \quad (1)$$

defined on a finite interval $[a, b]$ that is “regular” if

- The coefficient functions p , q , w and p' are all continuous on $[a, b]$
- $p(x) > 0$ and $w(x) > 0$ for all $x \in [a, b]$
- Boundary conditions are linear and homogeneous:

$$\begin{cases} \alpha_1 y'(a) + \beta_1 y(a) = 0 \\ \alpha_2 y'(b) + \beta_2 y(b) = 0 \end{cases} \quad \left(\begin{array}{l} \alpha_1 \text{ or } \beta_1 \\ \alpha_2 \text{ or } \beta_2 \end{array} \text{ can be zero} \right)$$

- $w(x)$ is called the weight or density function

Since 2nd-order, homogeneous ODE

$$a_0(x)y'' + a_1(x)y' + [a_2(x) + f(\lambda)a_3(x)]y = 0$$

can be put into the form (1). Hence

$$p(x) = e^{\int \frac{a_1(x')}{a_0(x')} dx'}, \quad q(x) = \frac{a_2(x)}{a_0(x)}, \quad w(x) = \frac{a_3(x)}{a_0(x)}$$

Properties of SL System

Sturm–Liouville theory states that, for any regular Sturm–Liouville problem:

1. All the eigenvalues are real
2. Corresponding to each eigenvalue λ_n is a unique (up to constant multiple) eigenfunction $\phi_n(x)$ with exactly $n - 1$ zeros in $[a, b]$, called the n th fundamental solution.
3. The normalized eigenfunctions $\hat{\phi}_n(x)$ form an orthonormal basis under the w -weighted inner product.
4. Eigenfunctions: a series of linearly independent solutions

$$y'' + \lambda^2 y = 0 \text{ with } \begin{cases} y(0) = 0 \\ y(L) = 0 \end{cases} \implies y = A \cos(\lambda x) + B \sin(\lambda x)$$

$$y(L) = B \sin(\lambda L) = 0 \implies \sin(\lambda L) = 0 \implies \lambda L = n\pi, n = 1, 2, 3\ldots \implies \lambda_n = \frac{n\pi}{L} \text{ (Eigenvalues)}$$

$$\phi_n = \sin\left(\frac{n\pi}{L}x\right), n = 1, 2, 3\ldots \text{ (Eigenfunctions)}$$

Orthogonality of Eigenfunctions

Orthogonality of functions: functions $f_m(x)$ and $f_n(x)$ are said to be orthogonal with respect to a weighting function $w(x)$ over the interval (a, b) if

$$\int_a^b w(x) f_m(x) f_n(x) dx = \begin{cases} 0 & \text{if } n \neq m \\ \text{const}(\neq 0) & \text{if } n = m \end{cases}$$

Theorem: Eigenfunctions to a S-L system are orthogonal

$$\phi_n = \sin\left(\frac{n\pi}{l}x\right), n = 1, 2, 3\ldots \text{ (Eigenfunctions)}$$

Expansion of a function in a series of orthogonal functions

Any piecewise continuous function over the interval (a, b) can be expanded in an infinite series of weighted eigenfunctions of ANY S-L system.

If $\{\phi_n(x)\}$ represent the eigenfunction set of a S-L system:

$$f(x) \Big|_{x \in (a,b)} = \sum_{n=1}^{\infty} A_n \phi_n(x)$$

How to determine the coefficient A_n ?

Determine the coefficient A_n

$$f(x) = \sum_{n=1}^{\infty} A_n \phi_n(x) \implies f(x) [\phi_k(x) w(x)] = \sum_{n=1}^{\infty} A_n \phi_n(x) [\phi_k(x) w(x)]$$

$$\int_a^b f(x) \phi_k(x) w(x) dx = \int_a^b \sum_{n=1}^{\infty} A_n \phi_n(x) \phi_k(x) w(x) dx$$

$$= \sum_{n=1}^{\infty} A_n \int_a^b \phi_n(x) \phi_k(x) w(x) dx$$

$$= A_k \int_a^b \phi_k^2(x) w(x) dx$$

$$A_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n^2(x) w(x) dx}$$

Solution of “Spherical” Diffusion Problem

$$\sum_{n=1}^{\infty} B_n \frac{\sin(\lambda_n r)}{r} = C_0 \quad \lambda_n = \frac{n\pi}{R_0}, n = 1, 2, \dots \quad \text{How to determine } B_n?$$

Look at the S-L system:

$$\begin{aligned} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \lambda^2 R r^2 &= 0 & \frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + q(x)y &= -f(\lambda)w(x)y \\ \text{s.t. } \begin{cases} R'(0) = 0 \\ R(R_0) = 0 \end{cases} & & \text{s.t. } \begin{cases} \alpha_1 y'(a) + \beta_1 y(a) = 0 \\ \alpha_2 y'(b) + \beta_2 y(b) = 0 \end{cases} \end{aligned}$$

The weighting function is $w(x) = r^2$:

$$B_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n^2(x) w(x) dx} = \frac{\int_0^{R_0} C_0 \frac{\sin(\lambda_n r)}{r} r^2 dr}{\int_0^{R_0} \frac{\sin^2(\lambda_n r)}{r^2} r^2 dr} = C_0 \frac{\int_0^{R_0} r \sin(\lambda_n r) dr}{\int_0^{R_0} \sin^2(\lambda_n r) dr} = C_0 \frac{I_1}{I_2}$$

Evaluate I_1 and I_2

$$\begin{aligned}
 I_1 &= \int_0^{R_0} r \sin(\lambda_n r) dr \\
 &= -\frac{1}{\lambda_n} \int_0^{R_0} r d \cos(\lambda_n r) \\
 &= -\frac{1}{\lambda_n} \left\{ [r \cos(\lambda_n r)]_0^{R_0} - \int_0^{R_0} \cos(\lambda_n r) dr \right\} \\
 &= -\frac{1}{\lambda_n} \left\{ R_0 \cos(\lambda_n R_0) - \frac{1}{\lambda_n} \int_0^{R_0} d \sin(\lambda_n r) \right\} \\
 &= -\frac{1}{\lambda_n} \left\{ R_0 \cos(\lambda_n R_0) - \frac{1}{\lambda_n} \sin(\lambda_n R_0) \right\} \\
 &= -\frac{1}{n\pi/R_0} \left\{ R_0 \cos(n\pi) - \frac{1}{n\pi/R_0} \sin(n\pi) \right\} \\
 &= \frac{R_0^2 (-1)^{n+1}}{n\pi}
 \end{aligned}$$

$$\begin{aligned}
 I_2 &= \int_0^{R_0} \sin^2(\lambda_n r) dr \\
 &= \int_0^{R_0} \frac{1 - \cos(2\lambda_n r)}{2} dr \\
 &= \frac{1}{2} \int_0^{R_0} dr - \frac{1}{2} \int_0^{R_0} \cos(2\lambda_n r) dr \\
 &= \frac{R_0}{2} - \frac{1}{4\lambda_n} \int_0^{R_0} \cos(2\lambda_n r) d(2\lambda_n r) \\
 &= \frac{R_0}{2} - \frac{1}{4\lambda_n} \sin(2\lambda_n r) \Big|_0^{R_0} \\
 &= \frac{R_0}{2} - \frac{1}{4(n\pi/R_0)} \sin(2(n\pi/R_0) R_0) \\
 &= \frac{R_0}{2}
 \end{aligned}$$

$$B_n = C_0 \frac{I_1}{I_2} = C_0 \frac{\frac{R_0^2 (-1)^{n+1}}{n\pi}}{\frac{R_0}{2}} = (-1)^{n+1} \frac{2R_0 C_0}{n\pi}$$

Final Solution of “Spherical” Diffusion Problem

$$\frac{\partial C}{\partial t} = \frac{D}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial C}{\partial r} \right)$$

$$IC : C(r, t = 0) = C_0$$

$$BC1 : \left. \frac{\partial C}{\partial r} \right|_{r=0} = 0$$

$$BC2 : C(R, t) = 0$$

$$\begin{aligned} C(r, t) &= \sum_{n=1}^{\infty} B_n \frac{\sin(\lambda_n r)}{r} e^{-\lambda_n^2 D t} \\ &= \frac{2R_0 C_0}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \frac{\sin\left(\frac{n\pi r}{R_0}\right)}{r} e^{-\left(\frac{n\pi}{R_0}\right)^2 D t} \end{aligned}$$

Applicability of using SOV

Which PDE can be solved by the technique of separation of variables:

1. PDE should be linear and homogeneous
2. Both boundary conditions for one independent variable should be linear and homogeneous

Example of Diffusion-Only Transport

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

$$IC : C(x, 0) = g(x) = \begin{cases} C_0 & (0 < x < a) \\ 0 & (a < x < L) \end{cases}$$

$$BCs : \left. \frac{\partial C}{\partial x} \right|_{x=0} = \left. \frac{\partial C}{\partial x} \right|_{x=L} = 0$$

How to
determine the
diffusion
coefficient D?

Example of Diffusion-Only Transport

$$\begin{aligned}\text{Assume } C(x, t) = X(x)T(t) &\implies \frac{\partial(XT)}{\partial t} = D \frac{\partial^2(XT)}{\partial x^2} \\ &\implies XT' = DTX'' \\ &\implies \frac{T'}{DT} = \frac{X''}{X} = \pm\lambda^2 (\text{which sign?})\end{aligned}$$

$$\text{T part: } T' + \lambda^2 DT = 0 \implies T = c_1 e^{-\lambda^2 Dt}$$

$$\text{X part: } \frac{X''}{X} = -\lambda^2 \implies X'' + \lambda^2 X = 0$$

$$\implies X(x) = A \sin(\lambda x) + B \cos(\lambda x)$$

$$BC : X'(0) = 0 \implies X'(0) = \lambda A \cos(\lambda 0) - B \lambda \sin(\lambda 0) = 0$$

$$\implies A = 0$$

$$BC : X'(L) = 0 \implies -B \lambda \sin(\lambda L) = 0$$

$$\implies \lambda_n = \frac{n\pi}{L}, n = 0, 1, 2, \dots$$

$$C(x, t) = \sum_{n=0}^{\infty} X_n(x) T_n(t) = \sum_{n=0}^{\infty} A_n \cos(\lambda_n x) e^{-\lambda_n^2 Dt} \quad \lambda_n = \frac{n\pi}{L}, n = 0, 1, 2, \dots$$

Example of Diffusion-Only Transport

Apply IC to determine A_n from

$$g(x) = C(x, 0) = \sum_{n=0}^{\infty} A_n \cos(\lambda_n x)$$

$$A_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n^2(x) w(x) dx} = \frac{\int_0^L g(x) \cos(\lambda_n x) \cdot 1 \cdot dx}{\int_0^L \cos^2(\lambda_n x) \cdot 1 \cdot dx} = \frac{C_0 \int_0^a \cos(\lambda_n x) dx}{\int_0^L \cos^2(\lambda_n x) dx}$$

$$C(x, t) = C_0 \frac{a}{L} + \frac{2C_0}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{n\pi a}{L}\right) \cos\left(\frac{n\pi x}{L}\right) e^{-(\frac{n\pi}{L})^2 Dt}$$

Suppose $a = \frac{L}{2}$, and we evaluate at $x = \frac{L}{6}$:

$$C\left(\frac{L}{6}, t\right) = \frac{C_0}{2} + \frac{2C_0}{\pi} \left(\frac{\sqrt{3}}{2}\right) e^{-(\frac{\pi}{L})^2 Dt} + 0 + 0 + 0 - \frac{2C_0}{5\pi} \left(\frac{\sqrt{3}}{2}\right) e^{-25(\frac{\pi}{L})^2 Dt} + \dots$$

$$C\left(\frac{L}{6}, t\right) - \frac{C_0}{2} \sim \frac{2C_0}{\pi} \left(\frac{\sqrt{3}}{2}\right) e^{-(\frac{\pi}{L})^2 Dt}$$

$$\ln \left[C\left(\frac{L}{6}, t\right) - \frac{C_0}{2} \right] - \ln \left[\frac{\sqrt{3}C_0}{\pi} \right] \sim -\left(\frac{\pi}{L}\right)^2 Dt$$

Plot the LHS vs t
to determine D

Example of Heat Conduction Problem

$$\rho C_p \frac{\partial T}{\partial t} = k \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right)$$

$$IC : T(r, 0) = T_\infty$$

$$BC1 : \left. \frac{\partial T}{\partial r} \right|_{r=0} = 0$$

$$BC2 : T(R, t) = T_0$$

Can we solve it by SOV?

What if the BC is not homogeneous?

Nondimensionalization:

$$\theta = \frac{T - T_0}{T_\infty - T_0}$$

$$x = \frac{r}{R}$$

$$\tau = \frac{k}{\rho C_p R^2} t$$

$$\frac{\partial \theta}{\partial \tau} = \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial \theta}{\partial x} \right) \quad \text{s.t.} \quad \begin{cases} IC : & \theta(x, 0) = 1 \\ BC1 : & \left. \frac{\partial \theta}{\partial x} \right|_{x=0} = 0 \\ BC2 : & \theta(1, \tau) = 0 \end{cases}$$

Assume $\theta(x, t) = X(x)T(\tau)$ and substitute into PDE and get:

$$\frac{\partial(XT)}{\partial \tau} = \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial(XT)}{\partial x} \right) \implies \frac{T'}{T} = \frac{1}{xX} \frac{d}{dx} \left(x \frac{dX}{dx} \right) = -\lambda^2$$

Example of Heat Conduction Problem

$$\text{T part: } \frac{T'}{T} = -\lambda^2 \implies T(\tau) = c_1 e^{-\lambda^2 \tau}$$

$$\text{X part: } \frac{1}{xX} \frac{d}{dx} \left(x \frac{dX}{dx} \right) = -\lambda^2 \implies \frac{d}{dx} \left(x \frac{dX}{dx} \right) + \lambda^2 x X = 0 \text{ (Bessel ODE)}$$

$$\text{Solve: } X = A J_0(\lambda x) + B Y_0(\lambda x)$$

$$\text{BC1 : } X'(0) = 0 \implies B = 0 \implies X = A J_0(\lambda x)$$

$$\text{BC2 : } X(1) = 0 \implies A J_0(\lambda) = 0$$

$$A \neq 0 \implies J_0(\lambda) = 0$$

$$\lambda_n \text{ is the } n\text{-th root of } J_0(x) = 0$$

$$\begin{cases} T_n(\tau) = c_{1n} e^{-\lambda_n^2 \tau} \\ X_n(x) = A_n J_0(\lambda_n x) \\ \lambda_n \text{ is the } n\text{-th root of } J_0(x) = 0 \end{cases} \implies \begin{aligned} \theta(x, t) &= \sum_{n=1}^{\infty} T_n(\tau) X_n(x) \\ &= \sum_{n=1}^{\infty} C_n J_0(\lambda_n x) e^{-\lambda_n^2 \tau} \end{aligned}$$

Example of Heat Conduction Problem

Apply IC to determine C_n :

$$\theta(x, 0) = \sum_{n=1}^{\infty} C_n J_0(\lambda_n x) = 1 = f(x) \text{ with the weighting function } w(x) = x$$

$$C_n = \frac{\int_a^b f(x) \phi_n(x) w(x) dx}{\int_a^b \phi_n^2(x) w(x) dx} = \frac{\int_0^1 1 \cdot J_0(\lambda_n x) \cdot x \cdot dx}{\int_0^1 J_0^2(\lambda_n x) \cdot x \cdot dx} = \frac{\frac{J_1(\lambda_n)}{\lambda_n}}{\frac{J_1^2(\lambda_n)}{2}} = \frac{2}{\lambda_n J_1(\lambda_n)}$$

Announcements

Week 9 Homework

- Due Date: Week 10 class – 2:00 PM, April 1, 2026
- Given 3 questions, only required to do **2** questions

Mid-Term Project

- Final submission of all the files (“exam report”, ppt slides, codes) by 23:59, **April 3**, 2026